

TRIBHUVAN UNIVERSITY
Institute of Science and Technology

NET CENTRIC COMPUTING

Unit-wise Previous Questions for B.Sc. CSIT (CSC367) and BIT (BIT351)

**Chapter-wise separation of examination questions
with Very Very Important (VVI) questions highlighted**

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Coverage: CSIT Model Set, CSIT 2078-2081, BIT 2080-2081

How to Use This Question Bank

Questions are arranged according to the nine chapters in the prescribed course contents. Within each chapter, B.Sc. CSIT and BIT questions are presented separately and then grouped by examination year or paper. Similar-looking questions are retained because they show how Tribhuvan University varies wording across examinations.

VVI	Very Very Important: repeated across papers/programmes, central to the syllabus, or commonly asked as a long/programming question.
Regular	Important syllabus question that should also be prepared.

Important note:

The VVI designation is an exam-preparation priority based on recurrence and syllabus emphasis. It is not an official guarantee of future examination questions.

Contents

1. Unit 1: Language Preliminaries (8 Hrs.)
2. Unit 2: Introduction to ASP.NET (3 Hrs.)
3. Unit 3: HTTP and ASP.NET Core (3 Hrs.)
4. Unit 4: Creating ASP.NET Core MVC Applications (10 Hrs.)
5. Unit 5: Working with Database (6 Hrs.)
6. Unit 6: State Management on ASP.NET Core Application (4 Hrs.)
7. Unit 7: Client-side Development in ASP.NET Core (4 Hrs.)
8. Unit 8: Securing ASP.NET Core Application (5 Hrs.)
9. Unit 9: Hosting and Deploying ASP.NET Core Application (2 Hrs.)

Unit 1: Language Preliminaries [8 Hrs.]

C# language constructs, inheritance, polymorphism, classes, delegates, generics, LINQ, exceptions, attributes, and asynchronous programming.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. Differentiate between abstract class, interface, and sealed class. When would you use each?

VVI 2. Write a program demonstrating an abstract class with abstract methods and a sealed class that prevents inheritance.

VVI 3. What are indexers in C#? How are they different from properties? Explain with an example.

4. Explain the async/await pattern in C#. Why is asynchronous programming important in web applications? Provide a simple example.

5. Write a short note on File I/O operations in C#.

6. Differentiate between method hiding and method overriding.

CSIT 2081

VVI 1. Distinguish between collections and generics.

2. What are named and positional attribute parameters?

VVI 3. Write a program to create a user-defined exception when the user enters a subject name other than "C#".

4. Create a class named EMPLOYEE as a superclass and ENGINEER and DOCTOR as subclasses. Make suitable assumptions regarding their properties and methods.

VVI 5. Write a short note on polymorphism.

CSIT 2080

VVI 1. Explain the need for partial classes and sealed classes.

VVI 2. How are delegates related to events?

VVI 3. Differentiate between generic and non-generic collections.

VVI 4. Write a program to create a generic class containing a generic constructor, generic member variable, generic property, and generic method.

5. Write a short note on lambda expressions.

CSIT 2079

VVI 1. Write a program to demonstrate collections and generics.

2. Write a short note on lambda expressions.

VVI 3. Write a short note on delegates and events.

CSIT 2078

1. Differentiate between a structure and an enumeration.

VVI 2. Why is exception handling necessary? Illustrate with a user-defined exception.

3. What are named and positional attribute parameters?

4. Write a short note on LINQ.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. Differentiate between an error and an exception.

VVI 2. Write a program that accepts marks for five subjects and throws an exception when a mark is negative or exceeds 100.

VVI 3. What is the use of the base keyword? Give an example.

BIT 2081 - Paper B

VVI 1. Define indexers.

VVI 2. What is the use of the base keyword?

VVI 3. How can polymorphism be applied to improve code extensibility?

BIT 2080

VVI 1. Differentiate between collections and generics.

VVI 2. Why do we need an abstract class? Explain with an example.

Unit 2: Introduction to ASP.NET [3 Hrs.]

.NET, .NET Core, Mono, ASP.NET technologies, architecture, CLI, MSIL, CLR, and the build-run-test-deploy workflow.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. Differentiate between .NET, .NET Core, and Mono frameworks.

VVI 2. Explain the compilation and execution process of .NET applications covering CLI, MSIL, and CLR.

CSIT 2081

VVI 1. Describe the procedure for deploying a .NET Core application.

CSIT 2079

1. What is the purpose of ASP.NET Web Forms?
2. Differentiate between the .NET Framework and ASP.NET Framework.
3. How do you test .NET Core applications?

CSIT 2078

VVI 1. Explain the procedure for building and running .NET Core applications.

VVI 2. Describe the .NET architecture, design, and principles.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. Explain MSIL and CLR.

BIT 2081 - Paper B

1. What are the tasks of ASP.NET Web Forms?

BIT 2080

VVI 1. Explain MSIL and CLR.**Unit 3: HTTP and ASP.NET Core [3 Hrs.]**

HTTP, request-response format, common architectures, MVC pattern, ASP.NET Core architecture, projects, and conventions.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. Describe the ASP.NET Core architecture overview. How does it differ from traditional ASP.NET Framework architecture?

CSIT 2081

VVI 1. How does ASP.NET Core MVC provide a pattern-based way to build dynamic websites? Explain.

CSIT 2080

VVI 1. Describe the importance of the MVC pattern in designing web applications.

CSIT 2079

VVI 1. Define HTTP. Explain the MVC pattern.

CSIT 2078

VVI 1. Explain the HTTP request and response message formats with examples.**B. BIT Questions (BIT351)**

BIT 2081 - Paper A

VVI 1. Describe the MVC pattern in ASP.NET.

BIT 2081 - Paper B

VVI 1. Explain the MVC pattern of ASP.NET.

BIT 2080

VVI 1. Discuss the architecture of ASP.NET Core.**Unit 4: Creating ASP.NET Core MVC Applications [10 Hrs.]**

Environment setup, controllers, actions, action results, Razor views, Tag Helpers, model binding, validation, routing, Web API, JSON, DI, and IoC.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. What are the different Action Result types in ASP.NET Core MVC? Explain with examples.**VVI 2. Design an API controller that returns JSON for a Product entity (ProductId, Name, Price, Category) with GET, POST, PUT, and DELETE operations.****VVI 3. What is model binding in ASP.NET Core? Explain server-side model validation with data annotations.**

CSIT 2081

- VVI 1. How do you create a controller? Illustrate with an example.**
- VVI 2. Describe the procedure of rendering HTML with Razor using a suitable scenario.**
- VVI 3. Describe the life cycle of the Dependency Injection container.**

CSIT 2080

- VVI 1. How do you create a controller? Mention requirements for rendering HTML.**
- VVI 2. List some features of URL routing.**
- VVI 3. Mention the types of services managed by the IoC container and explain how they are registered.**
- VVI 4. What are Tag Helpers? What do Tag Helpers provide?**

CSIT 2079

- VVI 1. What do Dependency Injection and the IoC container mean? Explain.**
- VVI 2. What is a Tag Helper? Discuss Razor syntax with an example.**

CSIT 2078

- VVI 1. Define URL routing. What does URL routing mean?**

B. BIT Questions (BIT351)**BIT 2081 - Paper A**

- VVI 1. What do you mean by Dependency Injection and the IoC container?**
 - VVI 2. Describe the methods for creating a controller and action with an example.**
3. Define JSON.

BIT 2081 - Paper B

- VVI 1. How do you create controllers, actions, and action results? Explain.**
- VVI 2. How is HTML rendered using views? Explain.**

BIT 2080

- VVI 1. Define URL routing and explain its features.**
 - VVI 2. Do model binding and validation refer to the same concept? Justify.**
3. How do you integrate REST API calls? Describe.

Unit 5: Working with Database [6 Hrs.]

ADO.NET classes, Entity Framework Core, ORM, providers, data models, data context, and CRUD operations.

A. B.Sc. CSIT Questions (CSC367)**CSIT Model Set**

- VVI 1. Explain the roles of Connection, Command, Reader, and Adapter classes in ADO.NET. How do they work together to perform database operations?**

CSIT 2081

- VVI 1. List the advantages of Entity Framework over ADO.NET.**

VVI 2. Using ADO.NET, insert records into FACULTY(Course_Name, CourseID, No_of_Semester, Fee) and retrieve the fee of the record whose course name is CSIT.

CSIT 2080

VVI 1. Using Entity Framework Core on BAG(ID, Brand, Price, Date): update price to 2000 for records dated 2022; delete records dated 1981; retrieve brands having price greater than 500.

VVI 2. What is the role of the Adapter class in ADO.NET?

CSIT 2079

VVI 1. Write an ADO.NET application to read records from MOVIE(Id, Name, Genre) where Genre is "Comedy".

CSIT 2078

VVI 1. Using Entity Framework Core, insert EmployeeId=3, Name="John", Salary=12000 into Employee(EmployeeId, Name, Salary).

VVI 2. Define data model and data context.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. Why do we need Reader and Adapter classes?

VVI 2. Using ADO.NET on BOOK(Id, AuthorName, Price), delete books having price less than 500.

BIT 2081 - Paper B

VVI 1. List the advantages of Entity Framework over ADO.NET.

VVI 2. Given BOOK(ISBN, Name, Author, Price), write a program to update the price to 1000 where the author is John.

BIT 2080

VVI 1. Describe data models and data context.

VVI 2. What is the purpose of an Object-Relational Mapper? Describe.

Unit 6: State Management on ASP.NET Core Application [4 Hrs.]

Stateless HTTP, Session State, TempData, HttpContext, Cache, Cookies, Query Strings, and Hidden Fields.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. What is TempData? How is it different from Session State and ViewData? When would you use TempData?

VVI 2. Explain Cache as a server-side state management strategy. What are the benefits of caching in web applications?

CSIT 2081

VVI 1. Define state management. Explain the parts of state management on the client side.

VVI 2. Write a short note on Hidden Field.

CSIT 2080

VVI 1. How do you cache information of the client and what are its uses?

VVI 2. How do you manage state on stateless HTTP?

VVI 3. How and when do you create hidden fields?

CSIT 2079

VVI 1. Explain the uses of cookies.

VVI 2. Discuss server-side state management strategies.

CSIT 2078

VVI 1. What are the advantages of stateless HTTP in an ASP.NET Core application?

VVI 2. Give an example to manage Session State.

3. Write a short note on Query String.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. Define cookies. What are the uses of hidden fields?

VVI 2. What is the use of Session State?

BIT 2081 - Paper B

VVI 1. How do you manage state on stateless HTTP?

BIT 2080

VVI 1. When do you need a hidden field? Explain with an example.

Unit 7: Client-side Development in ASP.NET Core [4 Hrs.]

Client-side technologies, jQuery, forms and validation, Single Page Applications, Angular, and React.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. Compare Angular and React for building Single Page Applications with an ASP.NET Core backend. Discuss the advantages and disadvantages of each.

CSIT 2081

VVI 1. Write a jQuery program to validate a form with username, password, and phone number. All fields must be non-empty and the phone number must contain exactly 10 digits.

CSIT 2080

VVI 1. How do you validate a form using jQuery? Describe with suitable assumptions.

VVI 2. Write a short note on Single Page Application.

CSIT 2079

VVI 1. Design a client-side validation application using jQuery for taking student input in a Library Management System. Make suitable assumptions.

VVI 2. What does Single Page Application mean?

CSIT 2078

VVI 1. Why do we need to validate a form? How can React be used for this?**B. BIT Questions (BIT351)**

BIT 2081 - Paper A

VVI 1. How do you create a form using React? Illustrate with an example.**VVI 2. Why do we need to validate a form on the client side? Justify with an example.****VVI 3. Discuss Single Page Applications.**

BIT 2081 - Paper B

VVI 1. Why do we need to validate the form? Describe.**VVI 2. Create a form using React with suitable assumptions.**

BIT 2080

VVI 1. What are the significances of a Single Page Application?**VVI 2. Discuss the use of Angular with an example.**

3. How do you integrate REST API calls? Describe.

Unit 8: Securing ASP.NET Core Application [5 Hrs.]

Identity, authentication, authorization, roles, claims, policies, secure controllers/actions, XSS, SQL Injection, CSRF, and Open Redirect attacks.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. What is ASP.NET Core Identity? Explain the process of adding authentication to an application and configuring Identity services.

CSIT 2081

VVI 1. Distinguish between roles and policies.**VVI 2. Discuss open redirect attacks.**

CSIT 2080

VVI 1. Explain the chances of SQL injection attack with a scenario.**VVI 2. Define claims and policies in terms of authorization.****VVI 3. Write a short note on Cross-Site Request Forgery.**

CSIT 2079

VVI 1. What types of vulnerabilities can occur in web applications? Show the case of a Cross-Site Scripting attack.**VVI 2. How do you assign roles, claims, and policies? Explain with an example.****VVI 3. What types of problems lead to SQL injection attacks?**

CSIT 2078

VVI 1. Write a program to show the possibility of an SQL injection attack.**VVI 2. Define open redirect attacks.**

VVI 3. How can you prevent SQL injection attacks? Explain with an example.

VVI 4. Write a short note on Open Redirect Attack.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. Illustrate the concept of SQL injection with an example.

VVI 2. How do you secure controllers and action methods?

BIT 2081 - Paper B

VVI 1. How do you detect and prevent vulnerabilities in an ASP.NET Core application?

VVI 2. What are the purposes of adding authentication to applications? Explain.

BIT 2080

VVI 1. Illustrate SQL injection attack with an example and explain its prevention procedure.

VVI 2. Why do we need to define roles, policies, and claims? Explain.

Unit 9: Hosting and Deploying ASP.NET Core Application [2 Hrs.]

IIS, Nginx, Apache, ASP.NET Core Module, Kestrel, Docker, containerization, and Azure publishing.

A. B.Sc. CSIT Questions (CSC367)

CSIT Model Set

VVI 1. Discuss ASP.NET Core hosting models: IIS, Nginx, Kestrel, and Docker containerization.

VVI 2. What is the ASP.NET Core Module? How does it work with IIS to host ASP.NET Core applications?

VVI 3. Write a short note on publishing an ASP.NET Core application to Azure Cloud.

CSIT 2081

VVI 1. Describe the procedure of deploying a .NET Core application.

VVI 2. Explain the Kestrel web server in ASP.NET Core.

CSIT 2080

VVI 1. State the steps to deploy a .NET Core application.

CSIT 2079

VVI 1. Describe any two hosting models for an ASP.NET Core application.

VVI 2. Discuss any two types of ASP.NET hosting models.

VVI 3. Write a short note on containerization using Docker.

CSIT 2078

VVI 1. Explain the procedure for building, running, and deploying .NET Core applications.

VVI 2. Discuss Docker and containerization.

VVI 3. Explain the process to deploy an ASP.NET Core application.

B. BIT Questions (BIT351)

BIT 2081 - Paper A

VVI 1. List the uses of Docker.

BIT 2081 - Paper B

VVI 1. Describe any two hosting models for an ASP.NET Core application.

BIT 2080

VVI 1. How do you publish an application to Azure Cloud?

Final VVI Revision Checklist

- VVI Collections vs generics; generic class; custom exception handling.**
- VVI Abstract, sealed, and interface concepts; base keyword; polymorphism.**
- VVI CLI, MSIL, CLR, and compilation/execution of .NET applications.**
- VVI HTTP request-response format, MVC pattern, and ASP.NET Core architecture.**
- VVI Controllers, actions, action result types, Razor views, Tag Helpers, routing, model binding, and validation.**
- VVI Dependency Injection, IoC container, service registration, and service lifetimes.**
- VVI ADO.NET Connection/Command/Reader/Adapter and CRUD programming.**
- VVI Entity Framework Core, ORM, data model, DbContext, and CRUD programming.**
- VVI State management: Session, TempData, Cache, Cookies, Query Strings, and Hidden Fields.**
- VVI React form, jQuery validation, Angular/React comparison, and SPA.**
- VVI SQL Injection with prevention; roles, claims, policies; authentication and Identity.**
- VVI Hosting models, Kestrel, ASP.NET Core Module, Docker, deployment, and Azure publishing.**

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